

APS ESAF - Experiment Hazard Control Plan Report

Printed date: 08/08/2026

PEN: 15-IDCD-2026-17

ESAF ID: 289784 (GUP)

ID Start Date: 02/06/2026 08:00 AM

ID End Date: 02/17/2026 04:00 PM

Spokesperson: Schlossman

GUP ID: 1018531

Title: Complexation of Lanthanides to LBT Peptide Surfactants at the Water Surface

Spokesperson

The information on this hazard control plan is accurate and complete. All materials/samples to be used and hazards have been identified. All users are listed. Activities are restricted to the scope of work declared in the ESAF.

Name	Institution	Phone
Bikash Sapkota	University of Illinois at Chicago	312-996-3400

Materials Hazards

Material	Qty	Tox	Bio	Flam	Rad	Carcin	Corro	Oxid	Expl	Nan	Othe	Disp	Lab
2-(N-morpholino)ethanesulfonic acid	10 g	N	N	N	N	N	N	N	N	N	N	Y	Y
Acetone	4 L	Y	N	Y	N	N	N	N	N	N	N	Y	Y
Ammonium Persulfate	100 g	Y	N	N	N	N	N	Y	N	N	N	Y	Y
Erbium(III) Chloride	25 g	Y	N	N	N	N	N	Y	N	N	N	Y	Y
Europium (III) Chloride	25 g	Y	N	N	N	N	N	N	N	N	N	Y	Y
Hydrochloric Acid	100 mL	Y	N	N	N	N	Y	N	N	N	N	Y	Y
LBT Peptide	1 g	N	N	N	N	N	N	N	N	N	N	Y	Y
Lanthanum(III) Chloride	25 g	N	N	N	N	N	Y	N	N	N	N	Y	Y
Lutetium(III) Chloride	25 g	N	N	N	N	N	N	N	N	N	N	Y	Y
Methanol	4 L	Y	N	Y	N	N	N	N	N	N	N	Y	Y
Neodymium(III) Chloride	25 g	N	N	N	N	N	Y	N	N	N	N	Y	Y
Sodium Chloride	500 g	N	N	N	N	N	N	N	N	N	N	Y	Y
Sodium Hydroxide	500 g	Y	N	N	N	N	Y	N	N	N	N	Y	Y
Sulfuric Acid	2.5 L	Y	N	N	N	N	Y	N	N	N	N	Y	Y
Terbium(III) Chloride	25 g	Y	N	N	N	N	N	N	N	N	N	Y	Y

Beamline Laboratory Used

Start Date: 06-FEB-26

End Date: 17-FEB-26

Activity Description:

Stock aqueous solutions will be prepared at the users home institution and transferred to APS. Refrigerate will be used to keep the LBT peptide solution to store. Small Counter space will be used to dilute the LBT peptide solution, Lanthanide Chloride solutions and mixed them in appropriate manner. In some cases, the pH meter may be used to adjust the pH to 2 for reference samples.

Sulfuric acid bath will be prepared by adding ~18 g/L of ammonium persulfate inside the fume hood space of the chemistry laboratory. Some glassware like pasture pipettes and graduated pipetted may be cleaned in this acid bath. After each sample, the sample tray will be transferred to the chemistry laboratory, cleaned with acetone, methanol and ultrapure water. The glass trays will then be transferred to acid bath, soaked in it for at least half an hour before taking out. We will have multiple sets of similar glass trays, so another glass trays which has been soaked into sulfuric acid baths for more than half an hour will be cleaned with pure

water, dried with compressed inert nitrogen gas set up at the chemistry laboratory.

All the works concerning Acetone, methanol and sulfuric acid will be done in the fume hood. Proper PPE will be worn at all times.

Planned used of chemical fume hood:Yes

Planned use of non-Beamline supplied equipment:No

Equipment Hazards

No equipment information is provided at this time.

Experiment Description

Samples consisting of aqueous solution of different Lanthanide(III) Chlorides with Lanthanide Binding Tag (LBT) peptide surfactants will be poured into a glass tray. The glass tray will be placed with a sample cell compatible for Liquid surface scattering experiments. Poured samples will be given approximately half an hour to form an interfacial layer with surfactant-lanthanide complex and Surface EXAFS measurements will be taken in Grazing Incident mode. Though sample will be prepared prior to beamtime at the home university of the user, it will be mixed in the chemistry laboratory.

Attached File:

Experiment Set up

Exp Start Date:

Exp End Date:

Experiment Description: Samples will be prepared in the home university of the user and transported to the APS. Samples will be mixed into the right proportion in chemistry lab at APS prior to pouring into the glass tray in the experimental hut. The glass tray will have been placed into a sample cell originally designed and used for the liquid-liquid interface experiments at the same experimental sector. The liquid surface level will be adjusted by the Syringe pump provided by the beamline. The sample cell will be flushed with Helium gas and sufficient time will be waited for the monolayer deposition before starting the EXAFS experiments. The samples themselves are stable and do not present any significant hazard. Precaution will be taken while pouring the sample into the glass tray.

After the experiment, the aqueous solution will be collected as waste and discarded following the waste disposal procedure. The glass tray will then be cleaned in sulfuric acid bath in the chemistry lab under the fume hood. Precautions will be taken and the procedure will be followed while using acid bath. All the waste generated during the experiment including the waste acid will be taken care following the appropriate procedure.

Experiment Safety Requirements Summary

Eng. Controls	Proc.Controls	D. Rev.& E. Inspec.	PPE	Signs Labeling	Dosim. Monitoring	Safety Training
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Eng. Controls	Proc.Controls	D. Rev.& E. Inspec.	PPE	Signs Labeling	Dosim. Monitoring	Safety Training
(1.0) As determined by Beamline Design Review Process.	(1.0) As determined by Beamline Design Review Process. Eating or drinking is not allowed in areas where chemicals are being used. In case of emergency or spill dial 911.	(1.0) APS reviews for standard beamline operation already exist.	(1.0) Safety glasses with side shields and appropriate gloves for incidental use of common solvents.	(1.0) Proper labeling of chemical containers.	(1.0) Dosimeter badges are required for all personnel on the APS Experiment Hall Floor.	(1.0) APS101, ESH100U, GERT, ESH223, Sector Orientation
(6.4) Store in flammable liquid cabinet	(6.4) SOP for use of flammable materials may be required. See ESAF Comments for additional requirements. Keep away from ignition sources.	(6.4) Dependent on quantity and containment of material.	(6.4) Flame-Resistant (FR) laboratory coat may be required as determined by review. Additional PPE determined in Experiment Review. See ESAF Comments for additional requirements.	(6.4) Determined in Experiment Review. See ESAF Comments for additional requirements.	(6.4) Appropriate detector for gases. Determined in Experiment Review.	(6.4) Determined in Experiment Review. See ESAF Comments for additional requirements.
(6.2) Store in acid cabinet and secondary spill tray.	(6.2) SOP for use of corrosive materials may be required. See ESAF Comments for additional requirements.	(6.2) Dependent on quantity and containment of material.	(6.2) No exposed hands and arms. Additional PPE determined in Experiment Review. See ESAF Comments for additional requirements.	(6.2) Determined in Experiment Review. See ESAF Comments for additional requirements.	(6.2) Determined in Experiment Review. See ESAF Comments for additional requirements.	(6.2) Determined in Experiment Review. See ESAF Comments for additional requirements.
(6.3) Store in secondary spill tray	(6.3) SOP for use of toxic materials may be required. See ESAF Comments for additional requirements.	(6.3) Dependent on quantity, toxicity, and containment of material.	(6.3) No exposed hands and arms. Additional PPE determined in Experiment Review. See ESAF Comments for additional requirements.	(6.3) Determined in Experiment Review. See ESAF Comments for additional requirements.	(6.3) Determined in Experiment Review. See ESAF Comments for additional requirements.	(6.3) Determined in Experiment Review. See ESAF Comments for additional requirements.
(6.5) Store in secondary spill tray	(6.5) Do not store near flammable/combustible material or organic compounds. SOP for use of oxidizing materials may be required. See ESAF Comments for additional requirements.	(6.5) Dependent on quantity and containment of material.	(6.5) Additional PPE determined in Experiment Review. See ESAF Comments for additional requirements.	(6.5) Determined in Experiment Review. See ESAF Comments for additional requirements.	(6.5) Determined in Experiment Review. See ESAF Comments for additional requirements.	(6.5) Determined in Experiment Review. See ESAF Comments for additional requirements.
(6.0) Chemical Fume Hood. Eyewash Station. Safety Shower	(6.0) Determined in review of chemicals and process, and listed in SOP.	(6.0) ESAF Review.	(6.0) Proper clothing. Chemical goggles or safety glasses. Chemically resistant gloves (refer to MSDS or APS safety representative)	(6.0) Manufacturer Label or NFPA 704 Label	(6.0) Dependent on chemicals in use. Determined in Experiment Safety Review. See Comments for additional requirements.	(6.0) APS core training. Additional training may be required for specific chemicals. See Comments for additional requirements.

Beamline Comments

No Beamline comment is provided at this time.

APS Experiment Review Board Comments

This experiment Risk Hazard is: Medium

Date: 23-JAN-26 Badge: 304632 Verify completed APS core training for all on-site users. Training must be completed before starting work.

Date: 23-JAN-26 Badge: 304632 Use proper PPE (eye protection and impervious gloves) when working with hazardous

materials. The need for the use of PPE extends to the setup/maintenance/trouble shooting/disassembly of equipment containing or that may have contained hazardous materials.

All sample prep work with hazardous materials must be performed in the chemical fume hood or glovebox. Properly dispose of any generated waste. Waste may not be transported by users. Cleanup the fume hood/glovebox when work is complete. The use of a chemically compatible liner inside of the fume hood/glovebox is recommended to prevent contamination of surfaces.

- Date: 23-JAN-26 Badge: 304632 Use proper PPE (eye protection and impervious gloves) when working with samples. Properly identify/label the contents of any vials, bottles, or containers used in the lab for sample prep. Do not leave open chemicals or sample materials unattended. All containers must be labeled and kept closed and properly stored when not in immediate use. Keep work areas neat and organized. Decontaminate all work areas and used hardware and glassware when sample preparation is complete.
- Date: 23-JAN-26 Badge: 304632 Properly store flammables when not in use. No more than 5 gallons of flammable material is to be stored outside of a flammable storage cabinet at any given time. Properly store oxidizers when not in use. Oxidizers must be stored separately and away from flammable and combustible materials. Acids and bases must be stored separately. Use proper PPE (eye protection and impervious gloves) when working with corrosives.
- Date: 23-JAN-26 Badge: 304632 Wear PPE (eye protection and impervious gloves) for setup/maintenance, trouble shooting, disassembly, and operation of the flow cell/syringe pump.

Experiment Personnel and Training Due Dates

User Type	First Name	Last Name	Access End Date	Access	Gert	APS101	ESH100U	ESH1000	Sector	ESH223
On-site	Erik	Binter	02-SEP-26	OK	12/08/2027	12/08/2027	12/08/2027	NTREQ	12/12/2027	12/08/2026
On-site	Bikash	Sapkota	25-MAR-27	OK	04/12/2027	04/11/2027	04/11/2027	NTREQ	12/16/2027	04/18/2027
On-site	Mark	Schlossman	27-MAY-28	OK	05/11/2027	05/11/2027	Completed	05/22/2027	Completed	04/19/2027

APS ESAF - Authorization Checklist

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TO BE COMPLETED BY THE FLOOR COORDINATOR WHEN POSTING

Checklist Description	Confirmed
The materials list is accurate and complete for the current experiment.	
Laboratory usage has been verified and is appropriately marked on the ESAF.	
User list is accurate and complete, required training is up to date, and all users have been issued dosimeters.	
All equipment in use on this experiment, either supplied by the beamline or not, has been declared.	
Appropriate inspections/reviews of any equipment has been conducted. (Inspections can include DEEI/electrical, pressure, UV/Laser, etc.)	
Users are aware of the material and equipment hazards, particularly those called out in the ESAF comments.	
All appropriate hazard signage and/or SOPs are posted and PPE provided.	
Users are aware of the temperature limits of any heating device(s) that may be used in the experiment. All experimental work will be conducted within these limits.	
Users are aware they are restricted to the scope of work described in the ESAF and any out of scope modifications need to be approved before work can proceed.	

Additional Comments:

Items required to be completed before Authorization:

APS Authorization: _____ **Date:** _____

Version 8,4/1/2020

APS ESAF - Experiment Authorization Form

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Name	Institution	Signature	Date
Bikash Sapkota	University of Illinois at Chicago	_____	_____

Safety Review and Approval

The Hazard plan for this experiment has been reviewed and approved. A copy of the plan, which also describes the activities that have been reviewed and authorized, is posted on the experiment enclosure(s) and at other locations where work might be performed.

Point of Contact	Beamline Management	Date of Approval	APS	Date of Approval
BU, WEI	BU, WEI	02-FEB-26	SCHMIDT, ELIZABETH	23-JAN-26
Host				
WALWARK, DAVID J				

Materials Hazards

Chem: Corrosive, Flammable, Oxidizer, Toxic

Equipment Hazards

No equipment information is provided at this time.

Safeguards Verification

This experiment Risk Hazard is: Medium

All required controls, training and safeguards are in place to start the experiment. All users are listed. The experiment's scope of work and Hazard Control Plan are consistent with requirements stated in the update.

Organization	Name	Signature	Date
Beamline	Wei Bu	_____	_____
Posted By		_____	_____

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Beamline	Wei Bu	_____	_____
Posted By		_____	_____